

A Social Listening and NLP-based Framework for Brand Protection on Social Media

1. Introduction and Background

1.1. Problem Description

In the era of digital media, brands in Vietnam face increasing pressure from public interactions on social platforms. The rapid spread of information—especially negative content, rumors, or misinformation—can have immediate and wide-ranging impacts on brand image and customer trust.

Although commercial solutions like Brandwatch and Talkwalker [1], [2] support real-time collection and visualization of social data, these platforms remain limited in deep semantic analysis, multi-label intent classification, and incorporating a claim-verification pipeline with industry-grade evidence sources [9][12].

In addition, linguistic characteristics of Vietnamese—including complex morphological structure and the scarcity of high-quality labeled data—continue to challenge the effectiveness of NLP models for tasks such as sentiment analysis or intent recognition (e.g., complaints, questions, praise, rumors, or spam) [3]. Furthermore, research on applying generative AI to automatic summarization of Vietnamese comments and the generation of event reports for decision-making remains limited and lacks standardization [4], [5].

The urgency of the problem is underscored in communications-crisis management, where response time is a key factor directly affecting the extent of damage. Therefore, developing a system capable of automated data analysis, trend forecasting, and early warning via channels such as email or SMS—while integrating into internal operational workflows—is considered a promising solution that provides strategic advantage for organizations in controlling and protecting digital reputation.

1.2. Research objectives

The objective is to develop an optimized Vietnamese Social Listening system to monitor and protect brand reputation on social media. The system is designed to integrate the following technical components:

- Multi-label sentiment and intent classification, using Vietnamese transformer models such as PhoBERT and ViSoBERT [3], [6].
- Fake-news detection and claim verification about the brand: extract brand-related claims, retrieve evidence (official channels, regulators, verified platforms), perform NLI inference to conclude Support/Refute/Insufficient Evidence, with cited sources [9][10][12][13][16].
- Early-warning triggers via email/SMS upon detecting negative signals likely to precipitate a communications crisis.
- Automatic summarization and event report generation, applying generative AI models such as ViT5 and BARTpho to support real-time decision-making [4], [5].

Based on these objectives, the study poses the following core questions:

- How can Vietnamese NLP models be optimized to analyze sentiment and communicative intent in social media data for Social Listening?
- What methods are effective for detecting and tracking negative trends or rumors affecting brands on Vietnamese social media?
- How should the alerting mechanism and automated reporting system be designed and implemented for Vietnamese Social Listening data?

1.3. Scope of the Project

The project focuses on researching and developing a Vietnamese-specific Social Listening system aimed at protecting brand reputation in the context of social media. The scope includes:

- **Data:** Vietnamese data collected from popular social platforms such as Facebook and TikTok. The dataset includes comments, public posts, and news headlines related to specific brands or product categories.
- **Natural Language Processing (NLP):** Emphasis on Vietnamese pre-trained models such as PhoBERT [3], ViSoBERT [6], ViT5 [4], and BARTpho [5] to implement tasks including sentiment classification, multi-label intent recognition, and summarization.
- **Evidence Index & claim verification:** Build an Evidence Index comprising official brand channels, government portals, and official brand websites for cross-checking [14][15].
- **Negative event alerts:** Develop mechanisms for early detection and instant alerts via email/SMS upon detecting phrases, comment threads, or spikes in volume related to negative, inappropriate, or crisis-indicative content.

The project scope is designed to produce a prototype with practical applicability in enterprise environments while contributing to academic understanding of Vietnamese social data analysis.

2. Proposed Solution

2.1. Solution Description

The proposed solution is a Vietnamese social-media analytics and monitoring system operating in near real-time, aimed at brand protection and early risk detection. Core components include:

- **Crawler Module:** Collect public content from social networks and online news (Facebook, TikTok, YouTube, news sites, etc.).
- **Vietnamese NLP Processing Module:** Apply models such as PhoBERT and ViSoBERT [3], [6] to perform:
 - Sentiment classification (positive, negative, neutral)
 - Multi-label intent classification (complaint, question, praise, rumor, etc.)
- **Fake-News & Fact-Checking Pipeline:**
 - Entity/brand linking
 - Claim extraction
 - Retrieval (ElasticSearch)
 - NLI cross-encoder (PhoBERT/ ViSoBERT/ XLM-R) to label Support/ Refute/ Insufficient Evidence.
 - Evidence and alerting policies [9][12][16].
- **Early Warning:** When large volumes of negative content or potential crisis indicators are detected, the system sends alerts via email/SMS APIs to the responsible personnel.
- **Automatic Summarization and Reporting:** Use generative models such as ViT5 and BARTpho [4], [5] to generate summaries of negative content, supporting rapid decision-making.

The solution targets high automation and flexible integration into internal operations, effectively meeting the objectives outlined in Section 1.2.

2.2. Software Architecture (Overview)

The system adopts a modular client-server design with flexible scalability. Architectural overview:

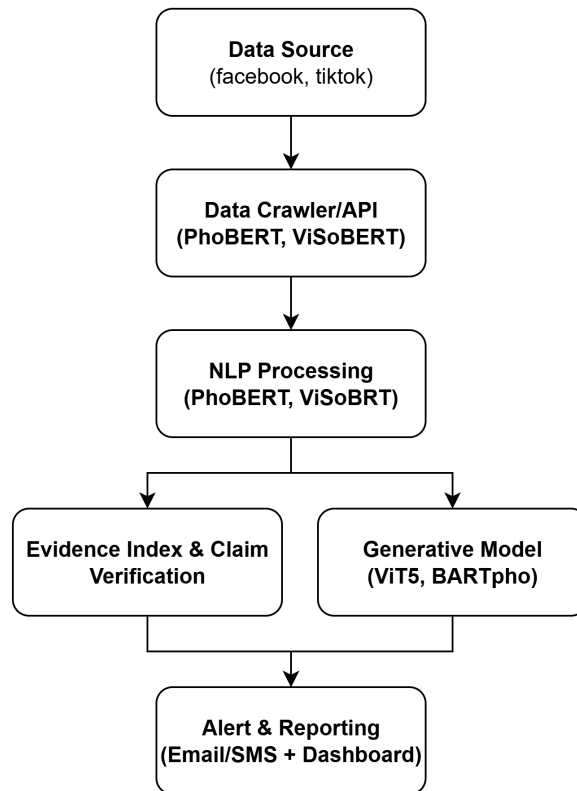


Figure 1. Architectural diagram illustrating data flow.

Interactions among components:

- The crawler ingests content from social media and passes it to the NLP pipeline for analysis.
- The pipeline indexes evidence and performs claim verification.
- The text-generation module produces summary reports.
- Results are displayed in the user interface and/or dispatched as alerts via APIs.

2.3. Technology and Tools Used

Technology choices are based on four criteria:

- Fit for Vietnamese language data
- Ease of integration with existing systems
- Processing efficiency and accuracy in real-world environments
- Scalability and long-term maintainability

TABLE I. Technologies and Tools Used

Component	Proposed Technology/Tool	Rationale
Programming language	Python	Strong support for NLP, AI, REST APIs
NLP framework	HuggingFace Transformers, PyTorch	Easy to deploy models like PhoBERT, ViT5
Crawler	Scrapy, Facebook Graph API, YouTube Data API	Suitable for collecting public data
Retrieval/Rerank	ElasticSearch, sentence-transformers (cross-encoder NLI)	Supports multimedia verification/web history [14][15]
Text generation	ViT5, BARTpho via Transformers	Vietnamese-appropriate models with strong performance
Alerting	Twilio API (SMS), SendGrid (email)	Easy integration, real-time alerting support
Database	PostgreSQL + ElasticSearch	Fast search, strong text-query support
Dashboard	Streamlit, Flask (REST API)	Lightweight, flexible user interface

With these choices, the system can handle large-scale social data while maintaining high accuracy in Vietnamese analysis—a language with complex grammar and expressive nuances.

3. Implementation Plan

3.1. Main development stages

The project is divided into five sequential phases to control progress and assess effectiveness at each step:

Phase 1: Requirements Gathering and Field Survey

- Interview stakeholders (businesses, communications experts) to identify specific needs.
- Review existing Social Listening tools and evaluate relevant Vietnamese NLP models.
- Define expected outputs and system evaluation criteria.

Phase 2: System Analysis and Design

- Design the software architecture (client-server, NLP modules, alerting, etc.).
- Build data-processing pipelines and alert-flow diagrams.

- Select appropriate models/modules for each function (sentiment, intent, generation, etc.).

Phase 3: Model and System Development

- Collect and clean Vietnamese datasets from social media and news.
- Implement NLP models (PhoBERT, ViSoBERT, ViT5, etc.) and train on target data.
- Build the data-collection system and fake-news pipeline with a Seed Evidence Index (brand-owned); fine-tune content-level fake-news detection on ReINTEL [11].

Phase 4: Testing and Tuning

- Test each module: classification, generation, alerting, dashboard.
- Evaluate accuracy (accuracy, F1-score), response speed, and early-detection capability.
- Optimize models and fix system issues based on internal pilot feedback.

Phase 5: Deployment and Evaluation

- Deploy a demo for pilot users.
- Gather real-world feedback; adjust alert logic and summarization.

3.2. Expected schedule (preliminary)

Phase	Estimated Time	Key Milestones
Requirements & Survey	Weeks 1–2	Define system requirements, evaluation metrics
Analysis & Design	Weeks 3–4	System diagrams, pipeline description, architecture docs
Model & System Development	Weeks 5–8	Complete NLP modules, crawling, pipeline, alerting
Testing & Optimization	Weeks 9–10	Model tuning, component integration
Demo Deployment & Evaluation	Weeks 11–12	Pilot demo with real data

3.3. Feasibility Assessment

The project scope fits short-term resources and timelines. Key challenges include: processing noisy, non-standard Vietnamese on social media; lack of high-quality multi-label annotated data; and near real-time responsiveness.

Data will be collected from Facebook and TikTok via public crawling/APIs, prioritizing high-engagement public content related to brands or sensitive topics, combined with keyword filtering. Tools: Selenium, BeautifulSoup, Scrapy. Data will be preprocessed, semi-automatically labeled, and added to the training set.

To mitigate data limitations and inconsistent Vietnamese usage, the proposal combines advanced preprocessing and existing datasets (VLSP [9], UIT-VSFC [10]). The system is modular, enabling iterative development and optimization; the alerting component updates continuously with small batches to detect negative trends early. The retrieval-plus-NLI approach to claim verification leverages the FEVER standard and Vietnamese datasets (ViFactCheck, ISE-DSC01), suitable for brand contexts [9][12][13].

Compute resources are ensured via low-cost cloud services (Google Colab Pro, HuggingFace Spaces). Open-source tools (Python, PyTorch, Elasticsearch, Streamlit, etc.) ensure feasibility, cost efficiency, and both academic and practical quality.

4. Expected Results

- Vietnamese Social Listening web app with capabilities to:
 - Collect public data from Facebook, TikTok, YouTube, and news sites.
 - Classify sentiment (positive, negative, neutral) and multi-label intent.
 - Detect fake news and verify claims with evidence citations (URL, title, publication date, excerpt) [9][12].
 - Maintain a per-brand Evidence Index (brand-owned).
 - Provide early alerts via email/SMS upon detecting negative signals.
 - Generate Vietnamese summaries and event reports.
- A specialized Vietnamese dataset with $\geq 10,000$ social-media samples labeled for sentiment and intent.
- Fine-tuned Vietnamese NLP models (PhoBERT, ViSoBERT, ViT5, BARTpho) packaged as reusable APIs.

5. Evaluation Plan

The system will be evaluated based on the following criteria:

- NLP Model Performance
 - Accuracy, Micro-F1, and Macro-F1 for sentiment, intent, and fake-news detection.
 - ROUGE-1, ROUGE-L for automatic summarization.
- System Performance
 - Data crawling speed (posts/comments per minute).
 - Latency from detecting a negative event to sending an alert (target ≤ 1 minute).

- Stability
 - Ability to operate continuously for 24 hours without critical failures.
 - Recovery after incidents (restart/reconnect).
- Usability
 - Pilot user satisfaction ($\geq 80\%$ satisfied).
 - Intuitive UI with easy access to reports and charts.

6. References

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